



A two-actor model for understanding user engagement with content creators: Applying social capital theory

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ABSTRACT

The emergence of video sharing platforms has given rise to the creation and consumption of tourism-related content. However, there is limited knowledge about the characteristics of content creators that enhance users' engagement with their content. The present study aims to fill this gap by examining creator characteristics and their impact on three tiers of user engagement. Tourism-related content, comprising 366 videos across six destinations, was extracted from YouTube using three social media analytic tools: VidIQ, TubeBuddy, and Social-Blade. The data were analyzed using PLS-SEM with SmartPLS 4.0. The findings reveal that channel subscribers positively influence user engagement at three levels – views, likes, and comments. However, a higher number of video uploads negatively impacts engagement. Furthermore, older videos tend to garner more views, but users' tendency to like the videos decreases over time. In addition, we extracted 23,993 comments and performed sentiment analysis on users' comments using Python-based VADER social media sentiment analysis tool. The compound-based sentiment analysis reveals that 59.5 percent of users show positive sentiments toward tourism-related content on YouTube while only 9.3 comments were negative, and 31.2 percent of sentiments remain neutral. Temporal analysis shows the rising trend in qualitative user engagement from 2010 to 2023, highlighting a growing interest in consuming and interacting with tourism-related content. This study discusses its theoretical contributions and managerial implications for content creators, destination managers, and advertising agencies.

1. Introduction

The tourism sector is widely regarded as one of the most information-intensive industries, as tourists typically seek a substantial amount of information before making decisions about visiting a particular destination (Alhemimah, 2023; Junaid et al., 2020). Travelers are keen to learn about a wide range of aspects, including tourist attractions, accommodations, dining options, restaurants, transportation, safety, local culture, and perhaps most importantly, the perspectives of fellow travelers who have visited these destinations. Travel vloggers and content creators play a crucial role in satisfying tourists' information needs by providing valuable insights about travel destinations through their online content (Silaban et al., 2022). Content creation has emerged as one of the most lucrative and fascinating sources of income, allowing content creators to produce content centered around their hobbies,

expertise, and interests while also making a living (Bhargava, 2022). In particular, the tourism sector is lending financial support to travel enthusiasts, enabling them to explore new places and generate income by creating content about the destinations they visit, the hotels they stay in, the restaurants they dine at, and the modes of transportation they use for commuting.

The process of content creation and distribution has been greatly facilitated by recent advances in social media networks and mobile devices (Calderón-Monge & Ramírez-Hurtado, 2021; Confetto et al., 2023; Khan, 2017; Royce Choi et al., 2023). Even solo travelers can easily create content using their smartphones and simply share it on platforms like YouTube, Instagram, and TikTok. This growing interest in content creation has given rise to an entirely new economic sector known as the "creator economy" (Rieder et al., 2023). Creator economy is an ecosystem of content creators, content users and advertisers

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facilitated by content sharing platforms like YouTube and Facebook (Bhargava, 2022). Within this novel economic ecosystem, content creators generate content for sharing on digital platforms, while their followers and subscribers consume this content to satisfy their various needs. Both content creators and digital platforms receive support from advertisers who promote their products and services during the consumption of this content. The interest of advertisers in utilizing content creators for the promotion of their products and services is steadily growing, given the continuous increase in digital advertising expenditure (Malhan & Dewani, 2023; Msallati, 2021). The emergence of content creators as key contenders in the marketing and advertising disciplines requires scholarly attention from the academic community. Particularly, user engagement with content creators is one of the most important aspects for academic debate as it impacts creators' monetization and advertisers' return on ad spend. The extant marketing literature attempts to understand online user engagement but it mainly focuses on measuring cognitive and emotional engagement while behavioral engagement has received less attention (Hollebeek et al., 2017; Choi et al., 2023). Existing research (Khan, 2017; Munaro et al., 2021) has investigated behavioral engagement with YouTube content, however a knowledge gap exists in the literature about the relationship between behavioral engagement and content creator's characteristics.

Particularly, the research on user engagement with content creators remains limited in the hospitality and tourism sector. For, instance, Abbasi et al. (2022) examine word of mouth and destination visit intention with travel Vloggers. However, they measure it using retrospective survey questionnaire which does not fully encapsulate actual user behavior. The present study adopts a more practical approach to measuring users' behavioral engagement with content published on YouTube in the form of views, likes and comments. Similarly, this study attempts to take pragmatic measures of creators' characteristics at two levels, i) creator profile which consist of number of subscribers and total video uploads, and ii) video characteristics which consist of video duration and total days of video publication. By studying the role of creator profile as a determinant of user engagement, this study provides a two-actor perspective towards our understanding of the social capital theory. The understanding of creator characteristics in engaging users has potential implications, however, the extant literature provides limited evidence on how such characteristics augment user engagement in the tourism context. The present study aims to address these gaps by examining the role of creator characteristics (profile and video) in fostering user engagement at three tiers; i) views (quantitative), ii) likes (quantitative), and iii) comments (qualitative) using the lens of social capital theory. The present study also extends social capital theory within the context of tourism content creation and consumption, recognizing that tourism content is produced and consumed within a social network. This investigation, which delves into the characteristics of content creators and the engagement of users within this social network, not only broadens the principles of social network theory but also addresses inconsistencies found in the existing literature regarding the causal role of social capital.

The present research advances the body of knowledge in the realm of tourism marketing, creator economy, social media marketing, destination marketing, and user behavior. Firstly, it extends social capital theory by integrating creator and video characteristics into a model that examines three tiers of user engagement. Secondly, it adopts a more realistic approach to understanding user engagement with tourism-related content by considering video views, likes, and comments, addressing a gap in the literature that often relies on retrospective methods. Thirdly, the present study employs sentiment analysis method to assess the depth and the substance of the two-way engagement between users and content creators. Finally, the study underscores the rising importance of the creator economy, emphasizing how content creators are becoming pivotal players by engaging users with their content, akin to traditional brands selling products and services.

2. Literature review

2.1. YouTube content in tourism

YouTube is one of the most well-known social media platforms (Coates et al., 2020) for video-sharing (Schwemmer & Ziewiecki, 2018). YouTube has gained recognition as a medium for tourism information search (Almeida-Santana & Moreno-Gil, 2017; Silaban et al., 2022). YouTube travel searches are centered around destinations, attractions, and travel plans. Recently, travel has represented the second-highest searched videos on YouTube (Roy et al., 2020). YouTube has transformed the method of content creation and consumption from inert one-way communication to co-creation (Royce Choi et al., 2023). YouTube platform allows for different levels of engagement between content creators and other users (Sjöblom et al., 2019). Content creators use digital platforms, such as YouTube to produce, disseminate, engage followers, and monetize their content (Bhargava, 2022). These creative artists and entrepreneurs typically offer unique information on a particular field of interest, such as travel (Valsesia et al., 2020). Hence, content creators often engage users and develop strong ties with them by satisfying their needs.

YouTube is the leading video sharing platform that influences travel behavior (Silaban et al., 2022) by allowing users to develop conversations and to engage with content (Calderón-Monge & Ramírez-Hurtado, 2021). This medium enables users to engage with travel content in different ways. For example, YouTube users can distribute travel information, and share travel experiences and opinions through comments, likes, views, as well as video and photo uploads during pre-, and post-trip stages (Kang & Schuett, 2013). Leaning on the foundations of the social capital theory and the COBRA framework, this study directed attention toward a topic that has not been given enough consideration by tourism researchers. More specifically, we aimed to examine two-actor relationships; the relationship between creator profile (antecedent) and user engagement (outcome). The present study explores the multifaceted nature of content creation and the primary factors causing users to engage on YouTube. This work advances the literature on social media engagement by developing a theoretical model (see Fig. 1) to gauge users' different engagement levels with creators' content on YouTube.

2.2. Social capital theory and creator characteristics

Despite the absence of a common definition of 'social capital' among scholars, the concept has been associated with the notion that resources are created within a network (Coleman, 1988). Generally, social capital refers to "the things created and used in social relationships and benefits coming from them." (Royce Choi et al., 2023, p. 4). Thus, the essence of social capital is the relationships between content creators and their networks, social media facilitates engagement via two-way communication between personalities and media users (Khan, 2017). Interesting to note that there has been inconsistency in the literature as to whether social capital is a cause or effect of online use (Williams, 2006). For example, Ellison et al. (2007) considered social capital as a cause. On the other hand, Putnam (2000) indicated that social capital is associated with reciprocity, denoting it is both a cause and effect. To better understand the social capital theory, it is necessary to examine the user engagement as an outcome since it reflects users' favorable and chosen actions (van Doorn et al., 2010), in response to creators' content in the form of liking, viewing, and commenting behaviors (Khan, 2017). User engagement behavior is, therefore, critical to creators' success. As such, we adopted the viewpoint of Putnam (2000) and we argued that YouTube is particularly appropriate for the nurturing of individual-level social capital as a result of enduring connectedness within the network that is best achieved by sustaining current social connections and facilitating the creation of new ones. Hence, social capital involves both the network and the impact of the network.

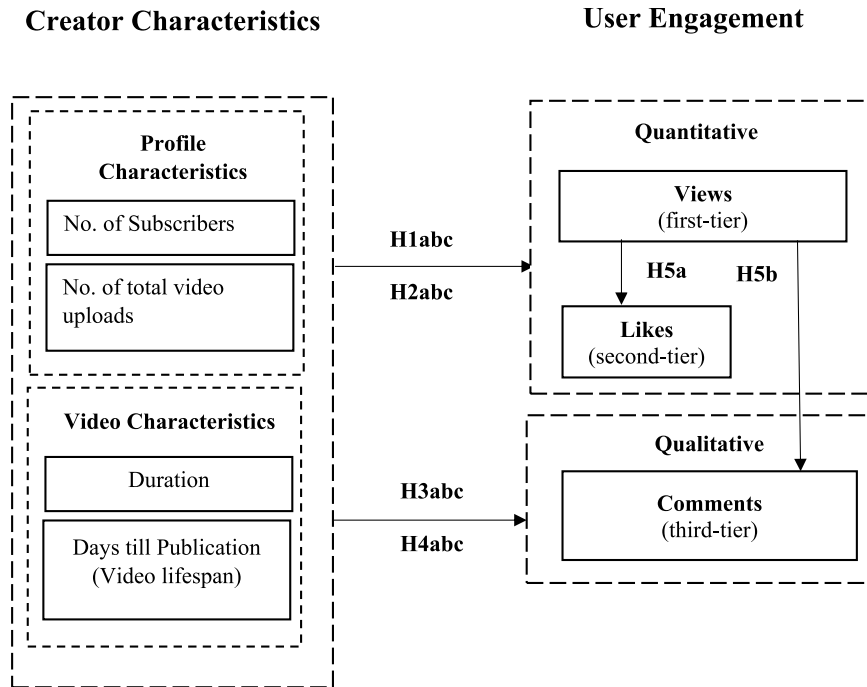


Fig. 1. Theoretical model.

Social capital has received attention from social media researchers interested in tourism. For example, [Musavengane and Kloppers \(2020\)](#) categorized social capital into two elements; rational and cognitive. Additionally, given its importance in social media settings, [Kim et al. \(2019\)](#) examined bonding and bridging types of social capital. Our literature review suggested that there are different dimensions for social capital. Social capital refers to “the goodwill available to individuals or groups. Its source lies in the structure and content of the actor’s social relations. Its effects flow from the information, influence, and solidarity it makes available to the actor” ([Adler & Kwon, 2002](#), p. 23). Adler and Kwon’s definition highlighted two key dimensions-social structure and social relations. This conceptual definition is consistent with our view as it associates social capital with two-actors; individual (creator characteristics) and collective (user engagement), this in turn expand the network of users by creating new ties. Structural capital refers to the connections between users in a social network ([Wasko & Faraj, 2005](#)). A social group can be described by two structural categories – creators’ connectedness and users’ integration ([Kleijnen et al., 2009](#)). First, social connectedness is measured by the number of connections (e.g. subscribers) content creators have in their social networks (e.g. network size). Second, user integration is measured by the frequency with which users engage with other community members in their network. This in turn influences the spreadability and the growth of the network. Relational social capital, on the other hand, is rooted in enduring interpersonal relationships ([van den Hooff & Huysman, 2009](#)) and the strength or closeness of these relationships ([De Bruyn & Lilien, 2008](#)). When users of digital platforms develop more active relationships with other users ([Oh et al., 2017](#)), this is an indicative of users’ engagement and their willingness to establish lines of communication owing to their interests in the digital content ([Wang et al., 2017](#)).

2.3. User engagement

User engagement construct in digital platforms has received significant attention from researchers (e.g., [Cerveny et al., 2021](#)). User engagement refers to “the intensity of an individual’s participation in and connection with an organization’s offerings and/or organizational

activities initiated by either the customer or the organization” ([Vivek et al., 2012](#), p. 127). Popular YouTube videos draw a large number of comments, views, and likes, hence promoting additional users’ engagements. Past tourism studies on digital platforms examined the user engagement as multidimensional concept that is comprised of three dimensions: cognitive (rational), emotional (feelings), and behavioral (actions) ([Kanje et al., 2020](#); [Vinerean & Opreana, 2021](#)). [Shawky et al. \(2020\)](#) called for additional research using a multilevel perspective to understand tourists’ engagement with social media networks. While some scholars highlighted the importance of the cognitive and emotional dimensions for the conceptualization of user engagement ([Gao et al., 2020](#); [Lim et al., 2020](#)), however other scholars argued that the behavioral factor is the most important since it induces other users to ‘act’ ([Hollebeek et al., 2017](#)).

Actions in the form of likes, views and comments on digital mediums have been documented in the literature as manifestations of behavioral engagement ([Bonsón et al., 2015](#)).

Users on YouTube may desire to stay passive by merely consuming creators’ content, or exhibit an active role by involving in any form of action. [Muntinga et al. \(2011\)](#) suggested the user’s online brand-related activities framework (COBRA) comprises of a three-step continuum of user engagement in digital mediums. This continuum gauges the three tiers of engagement ranging from passive engagement (consumption) to active engagements (contribution and creation). At the beginning of the continuum is “consumption” (represents minimum-level of engagement), users, for example, merely “view” a video. At the middle of the continuum is “contribution” (represents medium level of engagement), users, for example, click on the “like” button. At the end of this continuum is “creation” (represents high-level of engagement), users, for example, write “comments”. Since users’ levels of engagement are different and vary from one continuum to another ([Shawky et al., 2020](#)), this offers some interesting avenues for future research. Our study is grounded in the COBRA framework, which is regarded as the most notable framework ([Msallati, 2021](#)), contributes to the extant literature by investigating the three levels of user engagement.

2.4. Hypotheses development

This study unearths creators and video characteristics for YouTube user engagement that has been conceptualized as active participation and passive content consumption. Creators of content invest time and effort in the hope of gaining popularity and building reputation, which leads to boosting their social capital. Creators invest greatly in social capital and brand building (Rieder et al., 2023). Building on the foundations of the COBRA framework and the social capital theory, this research aimed to draw attention toward a relatively ignored set of actors—the interactions between creator content and the three tiers of user engagement.

We viewed social capital as both antecedent and outcome of behaviors. The interaction between the two actors (creator-user) helps in building social capital (Küper & Krämer, 2021). This study suggested that the two components of social capital – structure and relation – determine two categories of creator content, namely creator characteristics and video characteristics (Feroz Khan & Vong, 2014). Creator content includes social and non-social capital. The social capital of a YouTube video creator is defined in terms of the number of subscribers. The non-social capital of a YouTube video creator is defined in terms of the number of video uploads. In addition, the duration and video lifespan after the published date contribute to popularity of the video (Cha et al., 2007). YouTube video characteristics, on the other hand, are measured by video duration and video lifespan – days till publication date.

For content creators, user engagement represents the most crucial performance measure (Giertz, 2022). Digital platforms like YouTube define the different behavioral manifestations of user engagement; this engagement can be in the form of viewing, liking, and commenting activities (Bozkurt et al., 2020) to enhance popularity of videos (Kaplan & Haenlein, 2010). Importantly, to improve our understanding of this complex association, this requires an investigation of two-actor perspective as the relationships between user engagement and creator content are interdependent. Social capital represents the sum of generated, accrued, and distributed resources within a social community that is used for their benefits. Not all measures of social capital have numeric basis. To measure social capital of a YouTube video, Faucher (2018) indicated that it is necessary to examine both forms of social capital, namely quantitative (views and likes) and qualitative (comments). Therefore, following Faucher (2018), we express social capital in terms of numeric and non-numeric values. A recent study examines user engagement with number of subscribers on an educational platform and found a strong association between both variables (Fedushko et al., 2020). Thus, we hypothesize that:

Hypothesis H1a. The number of subscribers positively influences the first-tier of user engagement (number of views – quantitative metric).

Hypothesis H1b. The number of subscribers positively influences the second-tier of user engagement (number of likes – quantitative metric).

Hypothesis H1c. The number of subscribers positively influences the third-tier of user engagement (comments – qualitative metric).

A study by Khan and Vong (2014) reported a negative relationship between total videos published and video virality in terms of views, likes, and comments. Therefore, we proposed the following three hypotheses:

Hypothesis H2a. The number of video uploads negatively influences the first-tier of user engagement (number of views – quantitative metric).

Hypothesis H2b. The number of video uploads negatively influences the second-tier of user engagement (number of likes – quantitative metric).

Hypothesis H2c. The number of video uploads negatively influences the third-tier of user engagement (comments – qualitative metric).

Seo et al. (2021) reported a positive relationship between video length and the number of user activities with the digital platform. Past research has suggested more research to understand the relationship between video duration and user engagement (Dart, 2020). Based on these arguments, we proposed the following:

Hypothesis H3a. Video duration positively influences the first-tier of user engagement (number of views – quantitative metric).

Hypothesis H3b. Video duration positively influences the second-tier of user engagement (number of likes – quantitative metric).

Hypothesis H3c. Video duration positively influences the third-tier of user engagement (comments – qualitative metric).

Khan and Vong (2014) found a negative relationship between video published date and likes, views, and comments. A recent study by Stogowski et al. (2022) on the other hand, indicated that even if the video publication date is available, there still remains some uncertainty over whether the video content reflects the latest information about the topic of interest. Based on the above arguments, we propose the following null hypotheses:

Hypothesis H4a. The video lifespan (days till publication date) enhances first-tier user engagement (number of views – quantitative metric).

Hypothesis H4b. The video lifespan (days till publication date) enhances second-tier user engagement (number of likes – quantitative metric).

Hypothesis H4c. The video lifespan (days till publication date) enhances third-tier user engagement (comments – non-quantitative metric).

YouTube video views indicate popularity (Khan, 2017) and users' satisfaction with the brand (Lim et al., 2020). This study draws on past claims and considers that the number of views lead to a higher degree of engagement in the forms of liking and commenting. Therefore, we hypothesize the following:

Hypothesis H5a. The first-tier user engagement (number of views – quantitative metric) positively influences second-tier user engagement (number of likes – quantitative metric).

Hypothesis H5b. The first-tier user engagement (number of views – quantitative metric) positively influences third-tier user engagement (comments – qualitative metric).

3. Research methodology

The present study gathers and analyzes travel and tourism related video content published on YouTube across six Gulf Cooperation Countries (GCC). GCC is an association of six countries in the middle east which consists of Bahrain, Oman, Kuwait, Kingdom of Saudi Arabia, Qatar and United Arab Emirates. GCC countries are famous tourist destinations in the world which are attracting millions of tourists across the globe. According to United Nations World Tourism Organization these GCC countries attracted 47.7 million tourists in 2022 (UNWTO, 2023). Such a huge number of inbound tourists makes these GCC countries suitable candidates for research on content creation in social media. Therefore, content published on YouTube channels pertaining to GCC countries is the sample for our study. We followed a judgmental sampling approach (Hussain et al., 2022; Malhotra et al., 2017) to purposefully select the video content that fits well with aims, objectives and scope of our study. The video selected for this study must fulfil the following criteria: content must be relevant to hospitality and tourism, content must be related to GCC countries, content must be shared as video not as YouTube shorts, title must be relevant to the content of the video, video must rank among top 20 videos on searched keywords (Tubebuddy gives a list of top 20 videos on each keyword), and video must be publicly available.

All the videos fulfilling the above judgmental sampling criteria were extracted from YouTube. To avoid any YouTube's algorithm biases, we employed social media analytical tools to track the rankings of videos on different keywords and hashtags. YouTube's algorithm may influence the visibility of video content based on users' location, history, and consumption behavior, however, social media analytic tools track video rankings based on overall engagement with video regardless of current users' location and history. In addition, we used a set of keywords with different combinations so that same videos may not appear each time. We used two social media analytic tools i.e., VidIQ, and TubeBuddy for video rank tracking on keywords and hashtags. These tools also provided a range of quantitative matrices for each video which we extracted to utilize for data analysis purposes. We collected this data during the months of February, March and April 2023. For each country we started the video search with generic keywords e.g., "tourism places in [country name]", "tourist destinations in []", "places to visit in []", and "tourist attractions in []". Then we tracked the ranking of videos with TubeBuddy for each keyword. TubeBuddy gives top 20 videos that rank on a particular keyword; therefore, we started preparing the list of top videos that appeared on the keywords. In addition, TubeBuddy and VidIQ also show the tags used by the content creators. We used those tags to further track the videos ranked on those tags. When someone clicks on the tag (keyword) YouTube gives all the videos that rank on the tag. This process gives a list of videos that were showing repeatedly on different keywords and tags, therefore, we keep the top videos for each country ranked on different keywords but publish content about travel and tourism. We stopped searching new videos once the keywords got saturated and the same videos started appearing on different keywords. This process yielded 366 videos pertaining to six tourist destination countries. The details are documented in [Table 1](#).

Table 1
Content creator and video characteristics.

Factor		Value	Percentage
Number of Videos each Tourist Destination	Bahrain	49	13.4
	Kingdom of Saudi Arabia	72	19.7
	Kuwait	47	12.8
	Oman	68	18.6
	Qatar	58	15.8
	United Arab Emirates	72	19.7
Content Creators' Gender	Male	204	55.73
	Female	100	27.32
	Both	62	16.93
Language used in Videos	English	260	71.03
	English + Other	09	02.45
	Hindi	32	08.74
	Arabic	27	07.37
	Urdu	22	06.01
	French	03	00.81
	German	03	00.81
	Filipino	03	00.81
Countries where YouTube Channels are Registered	Others	07	01.91
	United States of America	88	24.04
	United Kingdom	52	14.20
	India	46	12.56
	United Arab Emirates	23	06.28
	Germany	20	05.46
	Kingdom of Saudi Arabia	13	03.55
	Pakistan	9	02.45
	New Zealand	8	02.18
	Kuwait	7	01.91
	Qatar	7	01.91
	Bahrain	4	01.91
	Oman	3	00.81
	Others	86	23.49
	Total	366	100

The video matrices were then tracked for these top videos. We extracted data for a range of matrices related to each video including number of views, likes, comments, date of publication, duration of the video, gender of the content creator and language of the content. Similarly, we also extracted data about the YouTube channel that published videos related to travel and tourism. Here we used SocialBlade – another social media analytics tool to document the information related to number of subscribers, total videos published by the channel, and country of origin of the channel. We also did some manual calculations using Excel formulas for the data which were not directly available. In total, we analyzed 366 videos relating to tourist attractions in six GCC countries. [Table 1](#) indicates that most of the videos talk about tourist attractions in the Kingdom of Saudi Arabia and United Arab Emirates followed by Oman and Qatar. Unsurprisingly most of the content (56 percent) produced by male creators which is twice the content produced by female creators (27 percent). English remains the dominant language with 71 percent content published in English language due to its global acceptance. Hindi, Arabic and Urdu are the other languages which were commonly used in travel vlogs. The YouTube channel origin has the wider spread among all variables as 24 percent channels are registered in the United States of America, 14 percent in the United Kingdom, 13 percent in India, and 16 percent in GCC countries.

Qualitative user engagement was captured through comments on content creators' videos. For this purpose, we utilized the YouTube Data API V3 to extract comments. However, this API has some limitations regarding comment extraction: it only retrieves the top 100 comments per video, regardless of the actual number, and extracts only the main comment while it is unable to extract replies to the comments. Despite these limitations, YouTube Data API remains the most suitable tool for comment extraction. Out of 366 originally selected videos, we successfully extracted comments from 355 videos. Eleven videos were excluded due to three reasons: 1 video had no comments, 6 videos had comments disabled, and 4 videos were unavailable. Consequently, we were able to collect 28,403 comments from 355 videos. Before diving into analysis, we cleaned the data: URLs, HTML elements, emojis, hashtags, special symbols, and excessive whitespaces were removed. The cleaning process resulted in empty strings in 4410 comments which were removed, leaving us with 23,993 comments for final analysis.

The data collected through social media analytic tools represents users' real engagement behavior and their responses to content creators. As [Khan \(2017\)](#) and [Bonsón et al. \(2015\)](#) suggest that views, likes and comments are the components of behavioral engagement. Similarly, [Hollebeek et al. \(2017\)](#) assert that behavioral engagement is the most important type of engagement because it stimulates further engagement from potential users. However, the extant tourism literature measures users' behavioral engagement with self-reported questionnaires while our study adopts a more pragmatic approach to measure and analyze users' behavioral engagement with tourism-related content creators. Encapsulating users' behavioral engagement offers several advantages over survey questionnaires, interviews, focus groups and other data collection. Particularly, the data about users' response to content creators' video content is based on users' real reactions as some users may like the content while others may not. The users react to those content without any researchers' involvement resulting in a less biased response. The survey-based data may encounter common method bias or users may not properly read survey questions. Whereas data extracted via social media analytic tools is not vulnerable to common method bias. Similarly, measuring behavioral engagement with video matrices mitigates measurement errors and error variance.

4. Results

The descriptive statistics of data are presented in [Table 2](#). These descriptive statistics provide valuable insights into the distribution and characteristics of the data for each variable in the dataset. The statistics indicate that the channel and video statistics have a wider spread. For

Table 2

Descriptive statistics of video content.

Variable	Minimum	Maximum	Mean	S.D
Number of Subscribers	134	15100000.00	619944.2514	1444512.96
Number of Views	354	8468563.00	430679.7787	885659.28
Number of Likes	0	138000.00	7720.3389	14541.65
Number of Comments	0	24417.00	732.2377	1738.62
Video Duration (Minutes)	1.0	56.00	15.3166	9.85
Video Lifespan (Days)	79.00	4782.00	985.9372	738.13
Total Video Uploads on a Channel	1.0	85723.00	778.6421	4758.07

instance, the number of channel subscribers varies from 134 subscribers to 15.10 million subscribers. Similarly, the number of views of tourism related videos range from 354 views to 8.46 million views. The minimum duration of the video is 1 min, and the longest video is 56 min long. The standard deviation shows that data has a greater variability, therefore, we standardized data by taking natural logs for all the variables used in the study to normalize the data variability. The extant literature suggests that log natural is one of the most appropriate methods to standardize data for spread and variability (Usman et al., 2018).

Table 3 documents the descriptive statistics after taking the natural logs. The values presented in the table indicate that data has suitable spread and variability for further analysis. In addition, the values of skewness and kurtosis are well below the +3 and -3 which suggest normal distribution of the data (Hair et al., 2010).

Table 4 shows the correlation matrix and variance inflation factor values (VIF). The values of correlation and VIF meet the cut off criterion (Hair et al., 2010) which indicates that multicollinearity is not an issue for the present study.

The present study analyzed data using PLS-SEM with smartPLS 4.0. The current study run the model in two steps i) PLS-SEM algorithm which gives statistics of model fitness and R-square and ii) PLS-SEM with bootstrapping (5000 subsamples) option which gives path coefficients with statistical significance. The model fit indices SRMR = 0.026 and NFI 0.936 indicate that the proposed model fits well with the data. The results presented in Table 5 and shown in Fig. 2 indicate that number of channel subscribers enhance number of video views ($\beta = 0.726$, $p < 0.05$) produced by the content creators which extends support to hypothesis H1a. Similarly, subscriber count positively influence likes ($\beta = 0.348$, $p < 0.05$) and comments ($\beta = 0.320$, $p < 0.05$) which approve our hypotheses H1b and H1c respectively. It is quite understandable that content creators who have amassed a large number of subscribers through the production of high-quality content have an advantage when it comes to garnering higher levels of user engagement. The subscriber count not only enhances first-tier engagement (i.e., views) but also contributes to increasing second-tier (i.e., likes) and third-tier (i.e., comments) engagement with the content. The results support our hypotheses H2a, b and c as total video uploads on a YouTube channel

Table 3

Descriptive statistics of video content after taking natural log.

Variable	Minimum	Maximum	Mean	S.D	Skewness	Kurtosis
Subscribers	4.90	16.53	11.47	2.32	-0.32	-0.62
Views	5.87	15.95	11.88	1.55	-0.40	1.32
Likes	0	11.84	7.74	1.74	-0.54	1.31
Comments	0	10.10	5.35	1.77	-0.61	0.77
Duration	1	4.03	2.50	0.70	-0.74	1.13
Video Lifespan	4.37	8.47	6.58	0.84	-0.37	-0.75
Total Video Uploads	1	11.36	5.49	1.25	0.17	2.28

negatively influence video views ($\beta = -0.314$, $p < 0.05$), likes ($\beta = -0.181$, $p < 0.05$) and comments ($\beta = -0.169$, $p < 0.05$). These results suggest that users do not want to be ambushed by content, however, they prefer to receive new content after an adequate interval. The content creators uploading lot of videos on their channels experience decreased user engagement with their content.

hypothesis H3a is not accepted as the path coefficients do not significantly support the hypothesis. H3a proposed that the video duration significantly influences number of views, but the empirical evidence suggests that the duration does not contribute to decreasing/enhancing number of views. The plausible reason might be the interest of users in finding relevant content instead of the length of the video. Users may be looking for more information that resonates with their searched terms and may watch a video that perfectly relates to their interest regardless of the video duration. Whereas H3b and c find significant support from the empirical evidence which suggests that likes ($\beta = 0.087$, $p < 0.05$) and comments ($\beta = 0.197$, $p < 0.05$) to video content increase as the duration of video increases. It reveals that users are interested in knowing more information about travel/tourism destinations in a video therefore a reasonable video duration may suffice users' information needs.

We accept hypothesis H4a ($\beta = 0.230$, $p < 0.05$) which suggests that the older videos get more views than the newly published videos. This is quite understandable as newly published videos may only appear to channel subscribers while older videos may appear in other searches and grab more views. Surprisingly, the results of H4b ($\beta = -0.063$, $p < 0.05$) reveal that days of video publication are negatively related with number of likes a video receives overtime. It indicates that users' tendency to like a video gets lower as the video gets older. It can be argued that people tend to like videos which provide most recent information about tourist places. Moreover, the finding of H4c ($\beta = -0.008$, $p > 0.05$) was not supported as the relationship between video publication date is not related to the number of comments received. Finally, the empirical evidence reveals that first-tier engagement (i.e., views) enhances second-tier (i.e., likes) and third-tier engagement (i.e., comments). The path coefficients indicate that higher number of views lead to higher number of likes to a video ($\beta = 0.674$, $p < 0.05$) and number of comments ($\beta = 0.506$, $p < 0.05$). These engagement statistics support our hypothesis H5a and b. The adjusted R^2 values presented in the table indicate that the predictor variables collectively explain 40 percent variance in views, 79 percent variance in likes and 57 percent variance in comments.

One contribution of this study is to analyze the quality of engagement through sentiment analysis of comments. For this purpose, we used a Python-based open-source library called VADER Sentiment Analysis (Hutto & Gilbert, 2014). VADER (Valence Aware Dictionary and sEntiment Reasoner) is a lexicon and rule-based sentiment analysis tool which is a specialized tool designed to perform sentiment analysis of social media text. Unlike formal/academic text, social media text does not follow formal rules of language and may include slang, emojis and combination of formal and informal language. VADER incorporates a sentiment lexicon containing words and phrases labeled with their sentiment intensity (positive, neutral, or negative) and it also assigns a compound score to each sentiment ranging from -1 (extremely negative) to +1 (extremely positive). Unlike other sentiment analysis tools, VADER calculates a compound score that captures the overall sentiment within an expression. Even if a comment contains positive, neutral, and negative words, VADER assigns individual scores to each sentiment and then combines them to generate a single, comprehensive score. It then uses this score to categorize the entire expression as positive, neutral, or negative. To perform sentiment analysis, we extracted and analyzed 23, 993 comments from 355 videos.

The quantitative analysis presented in the previous sections established that quantitative engagement metrics are valuable, but they have some limitations. To overcome these limitations, we used sentiment analysis which unlocks a deeper understanding of users' qualitative engagement with content creators. For instance, YouTube only provides

Table 4
Correlation matrix and variance inflation factor (VIF).

Variable	1	2	3	4	5	6	7
1. Subscribers	1.0	1.001	1.102	1.101	1.607	2.496	2.496
2. Total Video Uploads	0.409*	1.0	1.151	1.051	1.532	1.698	1.698
3. Duration	0.190*	0.094*	1.0	1.004	1.166	1.174	1.174
4. Video Lifespan	−0.035	−0.011	−0.222*	1.0	1.101	1.19	1.19
5. Views	0.371*	0.091*	0.144*	0.081*	1.0	1.687	1.687
6. Likes	0.462*	0.084*	0.226*	−0.032	0.716*	1.0	1.664
7. Comments	0.420*	0.089*	0.261*	−0.027	0.582*	0.736*	1.0

Note: The values above the diagonal represent VIF.

Table 5
Results of PLS-SEM analysis: Standardized coefficients.

Hypothesis	Hypothesized Path	β	S.D	T-Value	Result
H1a	Subscribers → Views	0.726***	0.044	16.413	Supported
H1b	Subscribers → Likes	0.348***	0.034	10.377	Supported
H1c	Subscribers → Comments	0.320***	0.047	6.741	Supported
H2a	Total Videos → Views	−0.314***	0.052	6.054	Supported
H2b	Total Videos → Likes	−0.181***	0.033	5.541	Supported
H2c	Total Videos → Comments	−0.169**	0.057	2.996	Supported
H3a	Duration → Views	0.069	0.046	1.498	Not Supported
H3b	Duration → Likes	0.087**	0.036	2.44	Supported
H3c	Duration → Comments	0.197***	0.056	3.491	Supported
H4a	Video Lifespan → Views	0.230***	0.039	5.668	Supported
H4b	Video Lifespan → Likes	−0.063**	0.024	2.668	Supported
H4c	Video Lifespan → Comments	−0.008	0.042	0.199	Not Supported
H5a	Views → Likes	0.674***	0.044	15.199	Supported
H5b	Views → Comments	0.506***	0.053	9.623	Supported
Variable		R ²	Adjusted R ²		
Views		0.407	0.401		
Likes		0.796	0.793		
Comments		0.575	0.569		

counts of likes that a video receives but has recently eliminated count of the dislikes which hampers our ability to gauge overall positive and negative impression of the video. Sentiment analysis offers a solution by analyzing comments and categorizing them as positive, neutral, or negative. The sentiment analysis for the present study is presented in Fig. 3 which demonstrates that 59.5% of users expressed positive sentiments toward tourism content creators, with 31.2% remaining neutral and only 9.3% expressing negativity. These findings portray a positive picture, indicating that creators are fostering deeper and more positive engagement through their content.

VADER also provides the impact of sentiments by analyzing the intensity of the sentiments and assigning weight to each sentiment. For instance, ‘amazing’ is a sentiment which is classified as “strong positive”

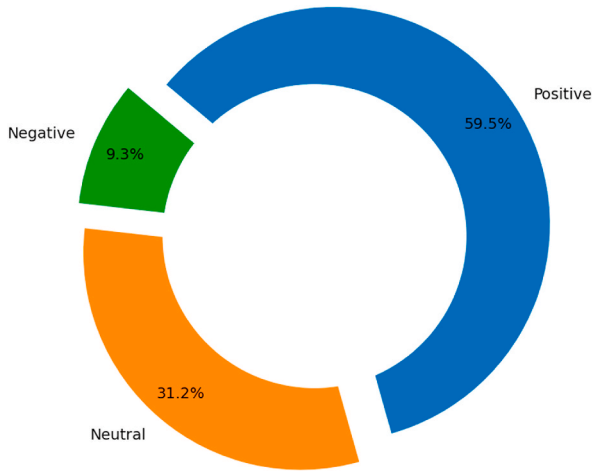


Fig. 3. Sentiment analysis of all comments.

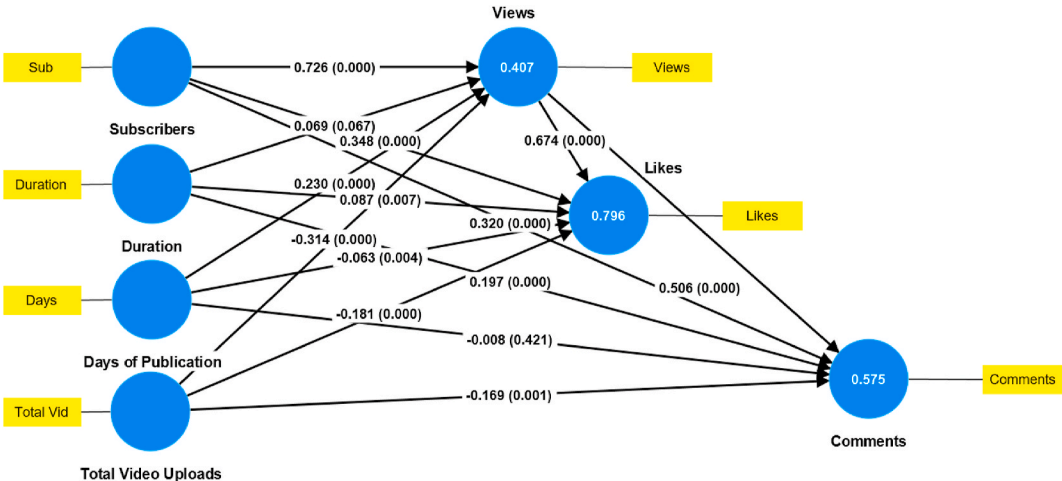


Fig. 2. PLS-SEM Model with path coefficients and p-value.

as compared to ‘good’ which is categorized as “weak positive”. The sentiment impact analysis shown in Fig. 4 reveals that the average impact of all the sentiments is positive while most of the positive sentiments have high impact which are skewed on the right side of the image. VADER assigned a numerical weight to each comment, for example this comment “I loved that u showed the amazing, beautiful side of [country name]” is categorized as strong positive with a weight of 0.9921 because it contains three positive words i.e., “love, amazing and beautiful”. In contrast, comments like “Please do not visit this Area [country name] now become very Dangerous” is classified as strong negative and receives -0.9764 impact score due to words dangerous and do not visit. While negative comments remained minimal in number, positive comments considerably outweighed them, suggesting an overall positive sentiment within the qualitative user engagement data. This impact analysis depicts a comprehensive picture of how users engage with tourism-related content on YouTube.

To delve deeper into sentiment analysis, we conducted sentiment assessments across four independent variables used in our theoretical model: number of subscribers, video duration, video lifespan, and total videos uploaded on the channel. Each independent variable was divided into five equal portions (20 percent increments): the top 20 percent, 21–40 percent, 41–60 percent, 61–80 percent, and bottom 20 percent. Dividing the data set in five quintiles is a structured way to analyze and compare users’ response pertaining to each variable using qualitative data like ours. It provides a middle ground for each category and helps to better understand the differences and commonalities among categories. After dividing the data in quintiles, we analyzed the sentiments associated with comments within each category. The results of these category-wise sentiment analyses are presented in Table 6. The statistics presented in table reveals that videos published on the channels with highest number of subscribers (i.e., Top 20 percent category) receive highest number of comments which supports the findings of our hypothesis H1c that number of subscribers positively influence qualitative engagement. For videos in this category, users showed 53 percent positive sentiments and 10 percent negative sentiments, while 37 percent of sentiments were neutral. Interestingly, other categories with fewer subscribers received relatively more positive sentiments than the highest subscriber category. This analysis uncovers insightful findings that having more comments is not valuable if users’ sentiments are not positive, as the top category received the least percentage of positive sentiments compared to all other categories.

The qualitative analysis of the second category pertaining to total number of video uploads on a channel depicts negative relation between number of video uploads and users’ sentiments. Users’ sentiments to such videos are comparatively less positive which indicates that users

may not want creators to upload too many videos. These findings support our earlier assertions that users do not want to be ambushed by content, however, they prefer to receive new content after an adequate interval. Video duration also validates the findings of our hypothesis H3c as the videos with longer durations receive more comments. Similar to subscriber count, longer videos receive relatively fewer positive sentiments from users as compared to shorter videos. Videos with 15 min or less duration are more favorably evaluated by consumers. Finally, the sentiment analysis reveals interesting findings regarding the video lifespan. The analysis depicts that recently uploaded videos receive more comments from users and the highest percentage of positive sentiments. The videos published within a year received 62 percent positive sentiments, which is the highest among all subcategories. The higher number of comments and positive sentiments received by recently uploaded videos indicates the temporal relevance of users’ qualitative engagement with the video content on YouTube.

In addition, we captured the temporal changes in qualitative engagement by measuring the number of comments content creators received over the years. Fig. 5 indicates that user engagement with creators’ content has sharply increased in recent years. Notably, engagement in 2022 was 100 percent higher than 2021 which represents a significant shift. A plausible reason for this exponential increase may be due to the 2022 FIFA football world in Qatar. While the 2022 FIFA World Cup in Qatar may have contributed to this surge in engagement by potentially increasing users’ interest in the GCC region, it’s important to note that the overall engagement level has also seen a continuous upward trend, highlighting a broader shift in user behavior.

5. Discussion

The present study investigates tourists’ engagement with travel and tourism-related video content published on YouTube. The current study bifurcates content creators’ profile in creator characteristics and video characteristics while encapsulating users’ behavioral engagement at three tiers. Though the extant literature attempts to understand users’ behavioral engagement with online content in the tourism sector, most of the literature relies on capturing engagement with retrospective methods of data collection. While our study follows a more pragmatic approach to explicating users’ behavioral engagement by their actual responses to video content published on YouTube. The findings suggest that existing subscriber count plays a key role in garnering views, likes and comments to video content produced by content creators. Therefore, the content creators may choose to avail YouTube’s channel promotion services which will enhance subscriber count on the channel. This might be more pertinent to the new content creators who do not have huge number of subscribers but create quality content. These findings resolved the inconsistencies documented in Fedushko et al. (2020) regarding the role of subscriber count in engaging users with digital content.

Our study presents some interesting findings that the higher number of video uploads on a YouTube channel decreases user engagement at all three levels. These results revert our attention to the very important issue that users do not want their timelines to be flooded with content but prefer a reasonable interval between posts. It may be argued that users prefer quality over quantity when consuming tourism-related content on YouTube. These findings corroborate with Khan’s (2017) assertion that users watch YouTube content for relaxing entertainment which is undermined by overly uploaded video content.

The present study also illustrates that video duration is an important indicator for garnering second and third-tier user engagement. It reveals that users show their engagement to those content which are slightly longer in length. It suggests that users want more details about a tourist destination in a video. The findings further elucidate that longer videos get more qualitative engagement in the form of comments from users. Arguably it happens due to the large amount of information presented in the video which stimulates users to participate (co-create) in the

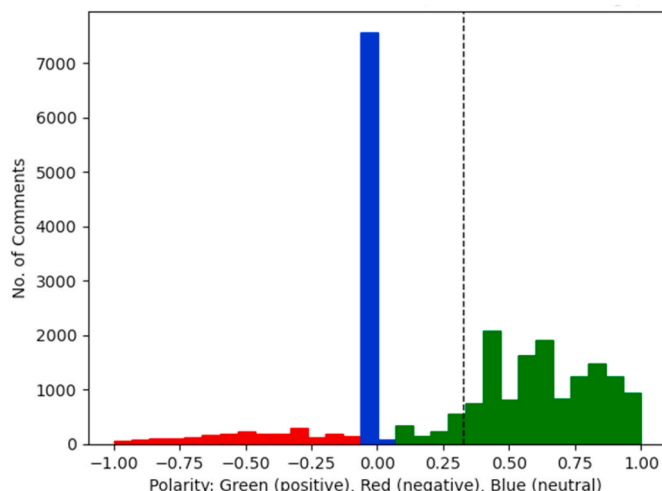


Fig. 4. Sentiment impact analysis (Dashed line indicates average value).

Table 6
Sentiment analysis across different categories.

Categorization relative to Subscriber Count	No. of Subscribers		No. of Comments	Sentiment Polarity (Values in percentage)		
	Minimum	Maximum		Positive	Neutral	Negative
Top 20 Percent (Highest subscribers)	768,000	15,100,000	5773	52.60	36.90	10.40
21–40 Percent	230,000	764,000	5568	61.30	29.10	09.70
41–60 Percent	63,700	224,000	5220	62.50	28.80	08.70
61–80 Percent	10,500	63,300	4113	61.00	30.50	08.60
Bottom 20 Percent (Lowest subscribers)	134	9990	3319	62.20	29.10	08.70
Categorization relative to Total No. of Video Uploads	Total No. of Video Uploads		No. of Comments	Sentiment Polarity (Values in percentage)		
	Minimum	Maximum		Positive	Neutral	Negative
Top 20 Percent (Most No. of Video Uploads)	617	85,723	5178	57.30	31.20	11.50
21–40 Percent	325	608	5247	54.10	37.00	08.90
41–60 Percent	168	325	4806	62.90	29.00	08.10
61–80 Percent	101	168	4700	62.50	29.90	07.70
Bottom 20 Percent (Least No. of Video Uploads)	01	99	4062	62.00	27.70	10.30
Categorization relative to Video Duration	Video Duration in Minutes		No. of Comments	Sentiment Polarity (Values in percentage)		
	Minimum	Maximum		Positive	Neutral	Negative
Top 20 Percent (Longest duration)	20	56	5537	57.80	31.90	10.30
21–40 Percent	15	20	5552	59.70	31.30	09.00
41–60 Percent	12	15	4674	60.20	31.40	08.40
61–80 Percent	8	11	4782	60.20	31.40	08.40
Bottom 20 Percent (Shortest duration)	1	8	3448	60.20	29.00	10.80
Categorization relative to Video's lifespan (Old vs newly published)	Video Lifespan in Days		No. of Comments	Sentiment Polarity (Values in percentage)		
	Minimum	Maximum		Positive	Neutral	Negative
Top 20 Percent (Longest lifespan – Older Videos)	1542	4782	4466	58.80	32.10	09.10
21–40 Percent	1139	1538	5024	60.30	31.20	08.40
41–60 Percent	610	1137	4743	58.50	31.90	09.60
61–80 Percent	322	587	4532	57.00	32.50	10.40
Bottom 20 Percent (Shortest lifespan – Recent Videos)	79	322	5228	62.40	28.40	09.20

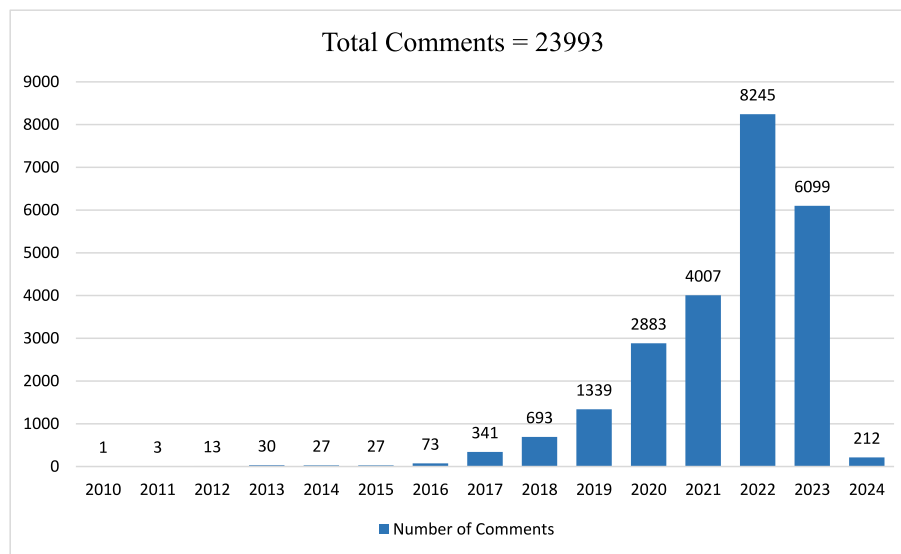


Fig. 5. Number of comments over the years.

conversation by appreciating content creators' efforts, sharing their personal experiences or asking for more details. These findings are in line with the extant literature which reveals that video length positively influence users' likes and comments tendency (Munaro et al., 2021). However, the insignificant impact of views indicates that users view a video regardless of the length but once they start viewing a video, they only engage with videos which provide more insights and reviews about a particular tourist attraction.

Our research also reveals crucial insights about the video lifespan (total days till publication of the video). For instance, older videos get more views but users' tendency to like the videos declines as the video gets older. It may suggest that users are looking for the latest information about the tourist destinations as they may be interested in booking hotel accommodation, transportation, and recent changes in the tourist

attractions.

The sentiment analysis of users' qualitative engagement offers interesting insights into users' expressions towards the video content. The findings show that most of the users positively evaluate tourism-related video content while a small percentage of users tend to express negativity towards the content. The users expressed gratitude, appreciation and contentment to social media content creators who spend their time, effort, and money in providing valuable information to the users. In addition, the qualitative analysis revealed motivations behind the users' engagement with tourism-related video content. We found that users show more engagement with content related to new destinations and content that offers entirely new information to users for facilitating their decision-making process. On the other hand, users also engaged with content created for destinations where they spent some time in the

past and the content instigated nostalgic feelings in them.

The present study offers several theoretical contributions to enhance the body of knowledge in the realm of tourism marketing, creator economy, social media marketing, destination marketing, and user behavior. First, the current study extends social capital theory by incorporating creator characteristics, video (content) characteristics and three tiers of user engagement in an integrated model. Users' participation in content creation in the form of comments provides deeper insights into social capital theory and highlights the importance of two-actor model in content creation and consumption in the context of tourism and hospitality. The application of social capital theory enabled the present study to encapsulate elements such as trust, reciprocity, and shared norms through sentiment analysis. Users showed confidence, love, and gratitude to content creators in their comments. While they reciprocated the content creators' efforts by subscribing to their channel, liking the content, and showing intentions to consume their content in the future. Second, the present research follows a more realistic approach to understanding users' behavioral engagement with tourism-related content by encapsulating video views, likes and comments while addressing a significant void in the literature which largely measures user engagement with survey methods. Third, the present study marks an initial attempt to investigate the impact of subscriber count, total video uploads, duration, and video lifespan with user engagement in the tourism context. Fourth, the current study analyzes tourism-related video content made for six countries by content creators of more than twenty countries which enriches academicians' understanding regarding content creation and consumption. Fifth, the present study marks the first attempt to analyze and compare users' sentiments towards tourism-related video content on YouTube across six countries. Sixth, the study also tracks the trend of users' temporal engagement with tourism-related content. Finally, the current study highlights the importance of creator economy as an emerging sector and how content creators are making their way into the novel sector by engaging customers with their content. In the creator economy, creators' content consumption is very much similar to other brands' sale of products and services. Simply put, our study stands apart from the extant literature by encapsulating users' real behaviors in the form of content consumption, engagement, and reciprocal interactions with content creators. Our application of social capital theory and analyses of the quantitative and qualitative engagement using creator characteristics and video metrics offer in-depth insights to better comprehend social media content creation and consumption in the realm of hospitality and tourism.

5.1. Managerial implications

The present study provides valuable insights for content creators, destination management organizations (e.g., brand and marketing managers), and advertising/marketing managers of hospitality and tourism services. Content creators can benefit the most by appropriately using creator and video characteristics to augment user engagement. For instance, the content creators' focus on gaining new subscribers is found to be worthwhile as higher subscriber count positively influences engagement at three tiers. However, our findings suggest that content creators may focus on quality content instead of quantity as the total number of video uploads on the channel negatively impacts user engagement. This observation aligns with the fact that users are inundated with an overwhelming volume of content on social media platforms, making it challenging for them to choose what to watch and engage with. Consequently, this inundation forces them to become highly selective in their consumption choices. Hence, tourism content creators may choose to limit the frequency of video uploads on their channels, as excessive posting can potentially undermine their reputation as creators of high-quality content. It is a common observation that users tend to associate sporadic content publication with quality, whereas an excess of content is often seen as an indicator of lower quality.

Interestingly, our study has unveiled a positive correlation between video duration and user engagement. This finding reinforces our argument that content creators should prioritize producing fewer videos with longer durations. Descriptive statistics indicate that the average video duration is approximately 15 min. Therefore, content creators should avoid producing videos shorter than this threshold. Longer videos also offer creators the opportunity to comprehensively cover various aspects of tourist destinations, including tourist attractions, booking information, accommodations, restaurants, cultural insights, safety tips, the cost of visiting, and, importantly, interactions with tourists on-site. If it is deemed necessary to include all the relevant information in the extended video, content creators can create sections using timestamps and labeling each section with a distinct title. This way, every user can conveniently skip information deemed less important to them. The content creators may also update their old videos by providing updated details because users intend to seek the latest information about a travel destination before they decide to visit it. Additionally, our study reveals a gender disparity among content creators, with 56 percent of videos being produced solely by males, 27 percent by females, and 17 percent by both genders. Female content creators may view this as an opportunity to increase their content production and establish a stronger presence in this field. The qualitative analysis of users' engagement reveals that users look for a diverse range of information that content creators overlook in their videos. Users want to know more about a lot of things before deciding to visit a tourist destination including hotel bookings, ticket prices, taxi riding, travel insurance and how to get to the place. Content creators may tackle this in two ways. First, content creators may provide brief information about such aspects within their content. Second, content creators may actively reply to the comments of users to answer questions.

Our study has established that tourism content creators can effectively engage users with their content, thereby creating opportunities for other businesses to market their products and services through these content creators. For example, marketing and brand managers at destination management organizations (DMOs) can leverage the influence of tourism content creators to promote inbound tourism. DMOs can identify well-known content creators with a substantial subscriber base and collaborate with them to produce content about their destinations, thus attracting tourists. Similarly, tourism and hospitality businesses such as hotels, restaurants, airlines, and tour operators can utilize tourism content creators for marketing their products and services. This becomes even more relevant since content creators can endorse these businesses specifically for the destination they are featuring in their content. As the content creators have the ability to reach a wide audience and influence their travel decisions.

6. Limitations of the study and future research direction

The present study has some limitations. For instance, the current study only focuses on YouTube as a content hosting platform. The future research may gather data from other platforms like Facebook, TikTok, and Instagram. The present study used only English keywords to search and gather data about tourism-related videos on YouTube. However, there exists a large number of languages and future research may search and analyze content published in other languages. YouTube is banned in some countries (e.g., China) so there may exist alternative video hosting platforms where content creators upload tourism related content and users refer to that content before making a final decision to visit a destination. The future research may search and gather data from those platforms to offer diverse insights. Exploring the distinction between videos created by males and females could provide valuable insights. Typically, males are expected to demonstrate a propensity towards objectivity and utility in their content, while females often exhibit a nuanced understanding of aesthetic appeal, particularly pertinent within the tourism context. Analyzing these differences can offer a deeper understanding of how gender perspectives shape content

creation in the field, thereby informing future strategies and approaches.

The present research tried to control YouTube's algorithm biases on content visibility and ranking by tracking video rankings using social media analytic tools. However, social media analytic tools may also have limitations in overcoming algorithm biases. Therefore, future research may identify some novel ways to control algorithm biases. In addition, the present study could not consider users' demographic and psychographic factors because YouTube does not provide any data about users' demographic and psychographic variables. Future research may look for ways to ascertain users' demographic and psychographic factors and investigate their influence on users' engagement behaviors.

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CRediT authorship contribution statement

Khalid Hussain: Conceptualization, Data curation, Formal analysis, Funding acquisition, Project administration, Writing – original draft, Writing – review & editing. **Khaldoon Nusair:** Conceptualization, Supervision, Writing – original draft, Writing – review & editing. **Muhammad Junaid:** Conceptualization, Methodology, Writing – original draft, Writing – review & editing. **Waqas Aman:** Data curation, Formal analysis, Methodology.

Declaration of competing interest

We do not have any conflict of interest to disclose.

Data availability

Data will be made available on request.

References

- Adler, P. S., & Kwon, S.-W. (2002). Social capital: Prospects for a new concept. *Academy of Management Review*, 27(1), 17–40. <https://doi.org/10.5465/amr.2002.5922314>
- Alhemimah, A. (2023). How COVID-19 impacts travel-health information seeking and tourists' travel intentions: A protection motivation theory-based model. *Journal of Destination Marketing & Management*, 27, Article 100757. <https://doi.org/10.1016/j.jdmm.2022.100757>
- Almeida-Santana, A., & Moreno-Gil, S. (2017). New trends in information search and their influence on destination loyalty: Digital destinations and relationship marketing. *Journal of Destination Marketing & Management*, 6(2), 150–161. <https://doi.org/10.1016/j.jdmm.2017.02.003>
- Bhargava, H. K. (2022). The creator economy: Managing ecosystem supply, revenue sharing, and platform design. *Management Science*, 68(7), 5233–5251. <https://doi.org/10.1287/mnsc.2021.4126>
- Bonsón, E., Royo, S., & Ratkai, M. (2015). Citizens' engagement on local governments' Facebook sites. An empirical analysis: The impact of different media and content types in Western Europe. *Government Information Quarterly*, 32(1), 52–62. <https://doi.org/10.1016/j.giq.2014.11.001>
- Bozkurt, S., Gligor, D. M., & Babin, B. J. (2020). The role of perceived firm social media interactivity in facilitating customer engagement behaviors. *European Journal of Marketing*, 55(4), 995–1022. <https://doi.org/10.1108/ejm-07-2019-0613>
- Calderón-Monge, E., & Ramírez-Hurtado, J. M. (2021). Measuring the consumer engagement related to social media: The case of franchising. *Electronic Commerce Research*, 22(4), 1249–1274. <https://doi.org/10.1007/s10660-021-09463-2>
- Cerveny, L. K., Derrien, M. M., Miller, A. B., & Meyer, C. (2021). Partnership and community engagement models for stewarding national scenic trails: A social-ecological Systems perspective. *Tourism Planning & Development*, 19(3), 204–226. <https://doi.org/10.1080/21568316.2021.1987314>
- Cha, M., Kwak, H., Rodriguez, P., Ahn, Y.-Y., & Moon, S. (2007). I tube, you tube, everybody tubes. In *Proceedings of the 7th ACM SIGCOMM conference on Internet measurement*. ACM. <https://doi.org/10.1145/1298306.1298309>
- Coates, A. E., Hardman, C. A., Halford, J. C. G., Christiansen, P., & Boyland, E. J. (2020). "It's just addictive people that make addictive videos": Children's understanding of and attitudes towards influencer marketing of food and beverages by YouTube video bloggers. *International Journal of Environmental Research and Public Health*, 17(2), 449. <https://doi.org/10.3390/ijerph17020449>
- Coleman, J. S. (1988). Social capital in the creation of human capital. *American Journal of Sociology*, 94, S95–S120. <https://doi.org/10.1086/228943>
- Confetto, M. G., Conte, F., Palazzo, M., & Siano, A. (2023). Digital destination branding: A framework to define and assess European DMOs practices. *Journal of Destination Marketing & Management*, 30, Article 100804. <https://doi.org/10.1016/j.jdmm.2023.100804>
- Dart, S. (2020). Khan-style video engagement in undergraduate engineering: Influence of video duration, content type and course. In *Proceedings of the 31st annual conference of the Australasian association for engineering education (AAEE 2020)*.
- De Bruyn, A., & Lilien, G. L. (2008). A multi-stage model of word-of-mouth influence through viral marketing. *International Journal of Research in Marketing*, 25(3), 151–163. <https://doi.org/10.1016/j.ijresmar.2008.03.004>
- Ellison, N. B., Steinfield, C., & Lampe, C. (2007). The benefits of Facebook "friends": social capital and college students' use of online social network sites. *Journal of Computer-Mediated Communication*, 12(4), 1143–1168. <https://doi.org/10.1111/j.1083-6101.2007.00367.x>
- Faucher, K. X. (2018). *Social capital online: Alienation and accumulation*. University of Westminster Press. <https://doi.org/10.16997/book16>
- Fedushko, S., Ustyianovych, T., Syerov, Y., & Peracek, T. (2020). User-engagement score and SLIs/SLOs/SLAs measurements correlation of E-business projects through big data analysis. *Applied Sciences*, 10(24), 9112. <https://doi.org/10.3390/app10249112>
- Feroz Khan, G., & Vong, S. (2014). Virality over YouTube: An empirical analysis. *Internet Research*, 24(5), 629–647. <https://doi.org/10.1108/intr-05-2013-0085>
- Gao, B. W., Jiang, J., & Tang, Y. (2020). The effect of blended learning platform and engagement on students' satisfaction—the case from the tourism management teaching. *Journal of Hospitality, Leisure, Sports and Tourism Education*, 27, Article 100272. <https://doi.org/10.1016/j.jhlste.2020.100272>
- Giertz, J. N. (2022). My friend, the content creator: Consequences of parasocial relationships for engagement in emerging social media [university goettingen repository]. <https://doi.org/10.53846/goediss-9352>
- Hair, J., Black, W., Babin, B., & Anderson, R. (2010). *Multivariate data analysis: A global perspective*. In *Multivariate data analysis: A global perspective* (7th ed., Vol. 7th).
- Hollebeek, L. D., Juric, B., & Tang, W. (2017). Virtual brand community engagement practices: A refined typology and model. *Journal of Services Marketing*, 31(3), 204–217. <https://doi.org/10.1108/jsm-01-2016-0006>
- Hussain, K., Junaid, M., Javed, M., Ali, M., & Iqbal, A. (2022). Consumer wisdom: A mindful connection between healthy food advertising and obesity prevention. *European Journal of Marketing*, 56(11), 3032–3053. <https://doi.org/10.1108/EJM-08-2021-0645>
- Hutto, C., & Gilbert, E. (2014). Vader: A parsimonious rule-based model for sentiment analysis of social media text. *Proceedings of the International AAAI Conference on Web and Social Media*, 8(1), 216–225.
- Junaid, M., Hussain, K., Akram, U., Asghar, M. M. M. M., Zafar, S., & Hou, F. (2020). Brand love: The emotional bridge between tourists' perceived value and well-being. *Asia Pacific Journal of Tourism Research*, 25(12), 1328–1341. <https://doi.org/10.1080/10941665.2020.1853577>
- Küper, A., & Krämer, N. C. (2021). Influencing factors for building social capital on live streaming websites. *Entertainment Computing*, 39, Article 100444. <https://doi.org/10.1016/j.entcom.2021.100444>
- Kang, M., & Schuett, M. A. (2013). Determinants of sharing travel experiences in social media. *Journal of Travel & Tourism Marketing*, 30(1–2), 93–107. <https://doi.org/10.1080/10548408.2013.751237>
- Kanje, P., Charles, G., Tumsifu, E., Mossberg, L., & Andersson, T. (2020). Customer engagement and eWOM in tourism. *Journal of Hospitality and Tourism Insights*, 3(3), 273–289. <https://doi.org/10.1108/jhti-04-2019-0074>
- Kaplan, A. M., & Haenlein, M. (2010). Users of the world, unite! The challenges and opportunities of Social Media. *Business Horizons*, 53(1), 59–68. <https://doi.org/10.1016/j.bushor.2009.09.003>
- Khan, M. L. (2017). Social media engagement: What motivates user participation and consumption on YouTube? *Computers in Human Behavior*, 66, 236–247.
- Kim, M. J., Preis, M. W., & Lee, C.-K. (2019). The effects of helping, self-expression, and enjoyment on social capital in social media: The moderating effect of avoidance attachment in the tourism context. *Behaviour & Information Technology*, 38(8), 760–781. <https://doi.org/10.1080/0144929X.2018.1552718>
- Kleijnen, M., Lievens, A., de Ruyter, K., & Wetzel, M. (2009). Knowledge creation through mobile social networks and its impact on intentions to use innovative mobile services. *Journal of Service Research*, 12(1), 15–35. <https://doi.org/10.1177/109467050933238>
- Lim, J. S., Choe, M.-J., Zhang, J., & Noh, G.-Y. (2020). The role of wishful identification, emotional engagement, and parasocial relationships in repeated viewing of live-streaming games: A social cognitive theory perspective. *Computers in Human Behavior*, 108, Article 106327. <https://doi.org/10.1016/j.chb.2020.106327>
- Malhan, M., & Dewani, P. P. (2023). Customer engagement on social networking sites: An experimental analysis in the tourism and hospitality sector. *Current Issues in Tourism*, 26(12), 1915–1940. <https://doi.org/10.1080/13683500.2022.2071685>
- Malhotra, N., Nunan, D., & Birks, D. (2017). *Marketing research: An applied approach*. Pearson.
- Msallati, A. (2021). Investigating the nexus between the types of advertising messages and customer engagement: Do customer involvement and generations matter? *Journal of Innovations in Digital Marketing*, 2(1), 1–13. <https://doi.org/10.51300/jidm-2020-31>
- Munaro, A. C., Hübner Barcelos, R., Francisco Maffezzoli, E. C., Santos Rodrigues, J. P., & Cabrera Paraiso, E. (2021). To engage or not engage? The features of video content on YouTube affecting digital consumer engagement. *Journal of Consumer Behaviour*, 20(5), 1336–1352.
- Muntinga, D. G., Moorman, M., & Smit, E. G. (2011). Introducing COBRAs. *International Journal of Advertising*, 30(1), 13–46. <https://doi.org/10.2501/ija-30-1-013-046>

- Musavengane, R., & Kloppers, R. (2020). Social capital: An investment towards community resilience in the collaborative natural resources management of community-based tourism schemes. *Tourism Management Perspectives*, 34, Article 100654. <https://doi.org/10.1016/j.tmp.2020.100654>
- Oh, C., Roumani, Y., Nwankpa, J. K., & Hu, H.-F. (2017). Beyond likes and tweets: Consumer engagement behavior and movie box office in social media. *Information & Management*, 54(1), 25–37. <https://doi.org/10.1016/j.im.2016.03.004>
- Putnam, R. D. (2000). Bowling alone. In *Proceedings of the 2000 ACM conference on Computer supported cooperative work*. ACM. <https://doi.org/10.1145/358916.361990>
- Rieder, B., Borra, E., Coromina, Ò., & Matamoros-Fernández, A. (2023). Making a living in the creator economy: A large-scale study of linking on YouTube. *Social Media + Society*, 9(2). <https://doi.org/10.1177/20563051231180628>
- Roy, S. K., Fatema, M. R., Islam, M. J., & Huda, M. N. (2020). Reviewing YouTube as a compelling tool for the promotion of tourism. *International Journal of Management and Accounting*, 96–104. <https://doi.org/10.34104/ijma.020.0960104>
- Royce Choi, J., Hong, S., & Kim, J. (2023). Does social capital matter to the millennials? Social capital and user engagements in online video platforms. *Telematics and Informatics*, 80, Article 101967. <https://doi.org/10.1016/j.tele.2023.101967>
- Schwemmer, C., & Ziewiecki, S. (2018). Social media sellout: The increasing role of product promotion on YouTube. *Social Media + Society*, 4(3). <https://doi.org/10.1177/2056305118786720>, 2056305118786720.
- Seo, K., Dodson, S., Harandi, N. M., Roberson, N., Fels, S., & Roll, I. (2021). Active learning with online video: The impact of learning context on engagement. *Computers & Education*, 165, Article 104132. <https://doi.org/10.1016/j.compedu.2021.104132>
- Shawky, S., Kubacki, K., Dietrich, T., & Weaven, S. (2020). A dynamic framework for managing customer engagement on social media. *Journal of Business Research*, 121, 567–577. <https://doi.org/10.1016/j.jbusres.2020.03.030>
- Silaban, P. H., Chen, W.-K., Nababan, T. S., Eunike, I. J., & Silalahi, A. D. K. (2022). How travel vlogs on YouTube influence consumer behavior: A use and gratification perspective and customer engagement (2022 *Human Behavior and Emerging Technologies*, 1–16. <https://doi.org/10.1155/2022/4432977>
- Sjöblom, M., Törhönen, M., Hamari, J., & Macey, J. (2019). The ingredients of Twitch streaming: Affordances of game streams. *Computers in Human Behavior*, 92, 20–28. <https://doi.org/10.1016/j.chb.2018.10.012>
- Stogowski, P., Antkowiak, L., Trzciński, R., Rogalska, M., Dulak, N. A., Anuszkiewicz, K., & Kloc, W. (2022). Content quality and audience engagement analysis of online videos for anterior lumbar interbody fusion. *World Neurosurgery*, 160, e636–e642. <https://doi.org/10.1016/j.wneu.2022.01.102>
- UNWTO. (2023). Global and regional tourism performance. <https://www.unwto.org/tourism-data/global-and-regional-tourism-performance>
- Usman, M., Zhang, J., Wang, F., Sun, J., & Makki, M. A. M. (2018). Gender diversity in compensation committees and CEO pay: Evidence from China. *Management Decision*, 56(5), 1065–1087.
- Valsesia, F., Proserpio, D., & Nunes, J. C. (2020). The positive effect of not following others on social media. *Journal of Marketing Research*, 57(6), 1152–1168. <https://doi.org/10.1177/0022243720915467>
- van den Hooff, B., & Huysman, M. (2009). Managing knowledge sharing: Emergent and engineering approaches. *Information & Management*, 46(1), 1–8. <https://doi.org/10.1016/j.im.2008.09.002>
- van Doorn, J., Lemon, K. N., Mittal, V., Nass, S., Pick, D., Pirner, P., & Verhoef, P. C. (2010). Customer engagement behavior: Theoretical foundations and research directions. *Journal of Service Research*, 13(3), 253–266. <https://doi.org/10.1177/1094670510375599>
- Vinerean, S., & Opreana, A. (2021). Measuring customer engagement in social media marketing: A higher-order model. *Journal of Theoretical and Applied Electronic Commerce Research*, 16(7), 2633–2654. <https://doi.org/10.3390/jtaer16070145>
- Vivek, S. D., Beatty, S. E., & Morgan, R. M. (2012). Customer engagement: Exploring customer relationships beyond purchase. *Journal of Marketing Theory and Practice*, 20(2), 122–146. <https://doi.org/10.2753/mtp1069-6679200201>
- Wang, R., Kim, J., Xiao, A., & Jung, Y. J. (2017). Networked narratives on Humans of New York: A content analysis of social media engagement on Facebook. *Computers in Human Behavior*, 66, 149–153. <https://doi.org/10.1016/j.chb.2016.09.042>
- Wasko, & Faraj. (2005). Why should I share? Examining social capital and knowledge contribution in electronic networks of Practice. *MIS Quarterly*, 29(1), 35. <https://doi.org/10.2307/25148667>
- Williams, D. (2006). On and off the 'net: Scales for social capital in an online era. *Journal of Computer-Mediated Communication*, 11(2), 593–628. <https://doi.org/10.1111/j.1083-6101.2006.00029.x>
- Zaib Abbasi, A., Hussain, K., Kaleem, T., Rasoolimanesh, S. M., Rasul, T., Ting, D. H., & Rather, R. A. (2022). Tourism promotion through vlog advertising and customer engagement behaviours of generation Z. *Current Issues in Tourism*, 1–20. <https://doi.org/10.1080/13683500.2022.2144156>

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